

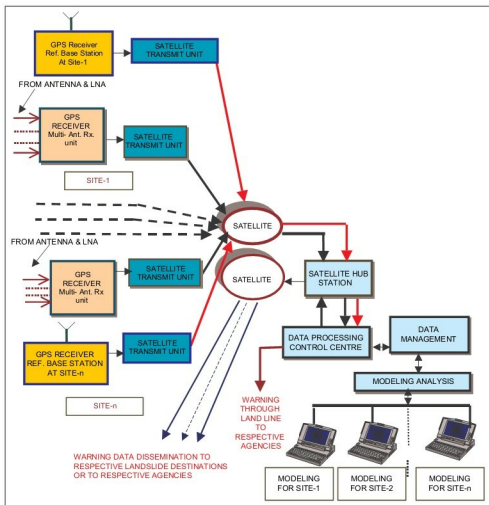


Fig. 8: Block diagram of landslide monitoring and alert system

be picked through the LAN.

Channel access. Two types of channel access are possible: single carrier shared by all the sites and separate carrier for each site.

Single-carrier access scheme. The TDMA technique can be implemented where each site is allotted a fixed time slot. The reference of time to maintain the correct time slot for each site is derived from the GPS receiver of each site. The main advantage of this scheme is reduced load on the satellite. Also, at the receive end, only one demodulator is required. But the TDMA technique is more complex and hence software implementation is costly.

Land deformation data update from each site in TDMA mode is shown in Fig. 7. As mentioned earlier, for high-accuracy measurements the

receiver could remain connected to each antenna for a set time period of 10 minutes (with the receiver sampling interval of 15 seconds) to allow it to acquire a number of epochs of data from an antenna before it switches to the next. Therefore the total time to collect data from 16 antennae would be 160 minutes (which means negligible switching time).

The data from each antenna (for 10 minutes) would be approximately 40 characters (each of 8 bits) long. The transmission time for a 40-character data burst (see Fig. 5) is 1.68 seconds, at the rate of 300 bps. The 40-character data consists of time, latitude, longitude and altitude. Therefore the total time to transfer data of 16 antennae from one site would be approximately 1.68×16 seconds plus guard bands be-

tween each transmission.

If guard band is 1 second, then total time = $(1.68 \times 16) + 16 \approx 43$ seconds (see Fig. 7). The total time for transferring data of the reference station from each site also would be 43 seconds.

Therefore the total time for data transfer from one site is 43 seconds + 43 seconds = 86 seconds.

If the number of sites for the entire country is 25, the total time of data transfer from all the 25 sites would be $86 \times 25 = 2150$ seconds = 36 minutes (approx.).

Therefore data from each site can be transferred to the central monitoring point (or hub station), at every 3 hours and 16 minutes ($160 + 36 = 196$ minutes) of interval.

Separate carrier for each site. This scheme is very simple because each site uses a separate carrier that is independent of the other sites. Also, there is no complexity at the receive end to recover data. However, its main disadvantages are the cost

of satellite carriers and the number of burst demodulators at the receive station (hub).

Data analysis

The position solutions of all the points are obtained based upon the raw GPS data from the monitoring zone. Standard coordinates of monitoring points from the traditional surveying techniques enable comparison of the deformation results. A threshold change of deformed-received data, compared with the ready model of each site, can be set. In case the threshold level is crossed, an alarm signal is sounded automatically to attract the attention of the operational staff for necessary action. ●

The author is a project manager-communication systems for disaster management, Space Application Centre, ISRO